that we measure should be optimized for black hole production. That is, any fiddling with these parameters of our universe should make it harder for black holes to form. Physicists have begun to investigate this prediction; at present there is no consensus on its validity. But even if Smolin's specific proposal turns out to be wrong, it does present yet another shape that the ultimate theory might take. The ultimate theory may, at first sight, appear to lack rigidity. We may find that it can describe a wealth of universes, most of which have no relevance to the one we inhabit. And moreover, we can imagine that this wealth of universes may be physically realized, leading to a multiverse—something that, at first sight, forever limits our predictive power. In fact, however, this discussion illustrates that an ultimate explanation can yet be achieved, so long as we grasp not only the ultimate laws but also their implications for cosmological evolution on an unexpectedly grand scale.

Undoubtedly, the cosmological implications of string/M-theory will be a major field of study well into the twenty-first century. Without accelerators capable of producing Planck-scale energies, we will increasingly have to rely on the cosmological accelerator of the big bang, and the relics it has left for us throughout the universe, for our experimental data. With luck and perseverance, we may finally be able to answer questions such as how the universe began, and why it has evolved to the form we behold in the heavens and on earth. There is, of course, much uncharted territory between where we are and where full answers to these fundamental questions lie. But the development of a quantum theory of gravity through superstring theory lends credence to the hope that we now possess theoretical tools for pushing into the vast regions of the unknown, and, no doubt after many a struggle, possibly emerging with answers to some of the deepest questions ever posed.

Part V: Unification in the Twenty-First Century